

Simulated Annealing

Simulated Annealing (SA) is a randomized algorithm, which approximates the global optimum of a function. It's called a randomized algorithm, because it employs a certain amount of randomness in its search and thus its output can vary for the same input.

The problem

We are given a function $E(s)$, which calculates the energy of the state s . We are tasked with finding the state s_{best} at which $E(s)$ is minimized. **SA** is suited for problems where the states are discrete and $E(s)$ has multiple local minima. We'll take the example of the [Travelling Salesman Problem \(TSP\)](#).

Travelling Salesman Problem (TSP)

You are given a set of nodes in 2 dimensional space. Each node is characterised by its x and y coordinates. Your task is to find an ordering of the nodes, which will minimise the distance to be travelled when visiting these nodes in that order.

Motivation

Annealing is a metallurgical process, wherein a material is heated up and allowed to cool, in order to allow the atoms inside to rearrange themselves in an arrangement with minimal internal energy, which in turn causes the material to have different properties. The state is the arrangement of atoms and the internal energy is the function being minimised. We can think of the original state of the atoms, as a local minima for its internal energy. To make the material rearrange its atoms, we need to motivate it to go across a region where its internal energy is not minimised in order to reach the global minima. This motivation is given by heating the material to a higher temperature.

Simulated annealing, literally, simulates this process. We start off with some random state (material) and set a high temperature (heat it up). Now, the algorithm is ready to accept states which have a higher energy than the current state, as it is motivated by the high temperature. This prevents the algorithm from getting stuck inside local minimas and move towards the global minima. As time progresses, the algorithm cools down and refuses the states with higher energy and moves into the closest minima it has found.